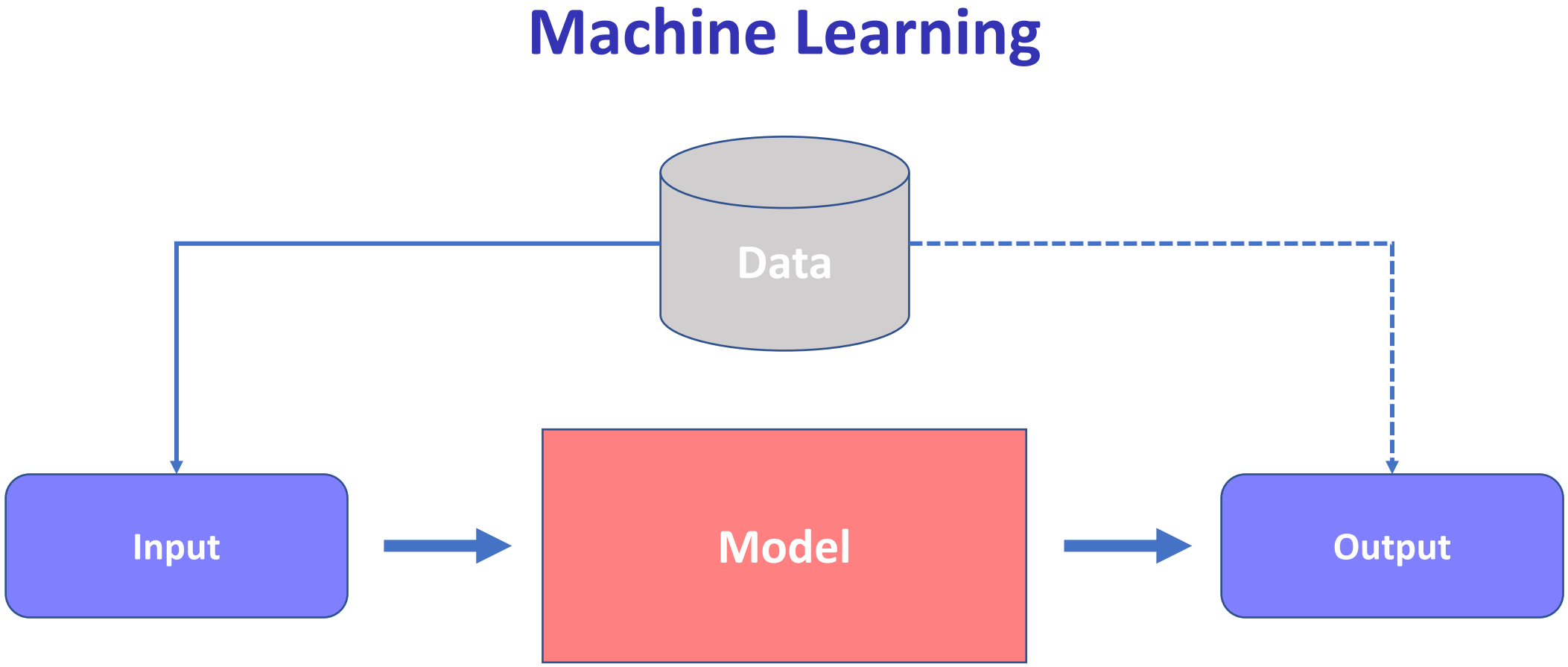
Generative Modeling

Machine Learning as Input-Output



Beyond standard deep networks

- Most common setup (e.g., AlexNet): Supervised Discriminative Learning
  - Supervised: Correct output (labels) are provided for a set of input examples
  - Discriminative: *Directly* model the correct output given the input
  - objects given image, pixel-level depth given image, sentiment given a text,
  - Most famous architectures are designed for a supervised discriminative task: AlexNet, Inception, ResNet, FCN, U-Net, LSTM, etc.

What is a generative model?

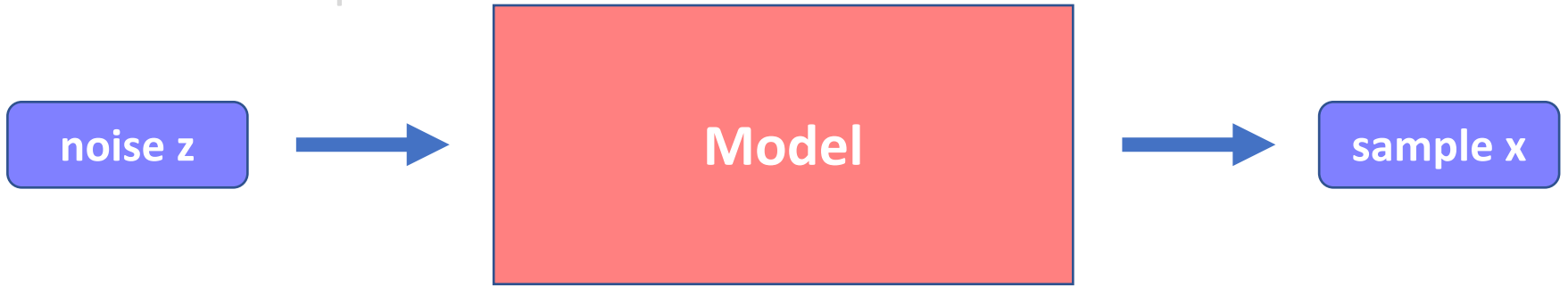
let us define it by its goals

Generative Models: Goals

1st goal: Sample generation

- Generate realistic samples (from the same distribution as training data, *i.e.*, P(**x**))
  - latent variable models
  - fully-observable models

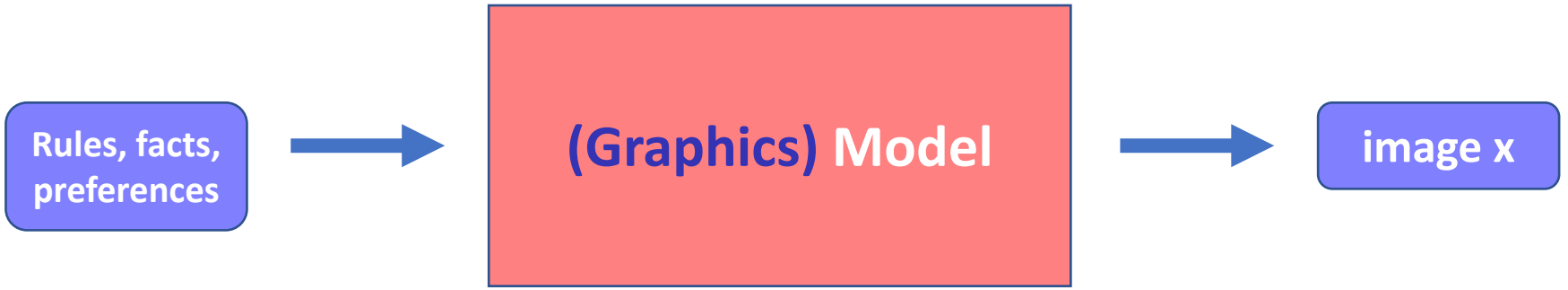
- Assign likelihood to samples



Generative Models: Goals

1st goal: Sample generation

- Generate realistic samples a concrete case
- Assign likelihood to samples

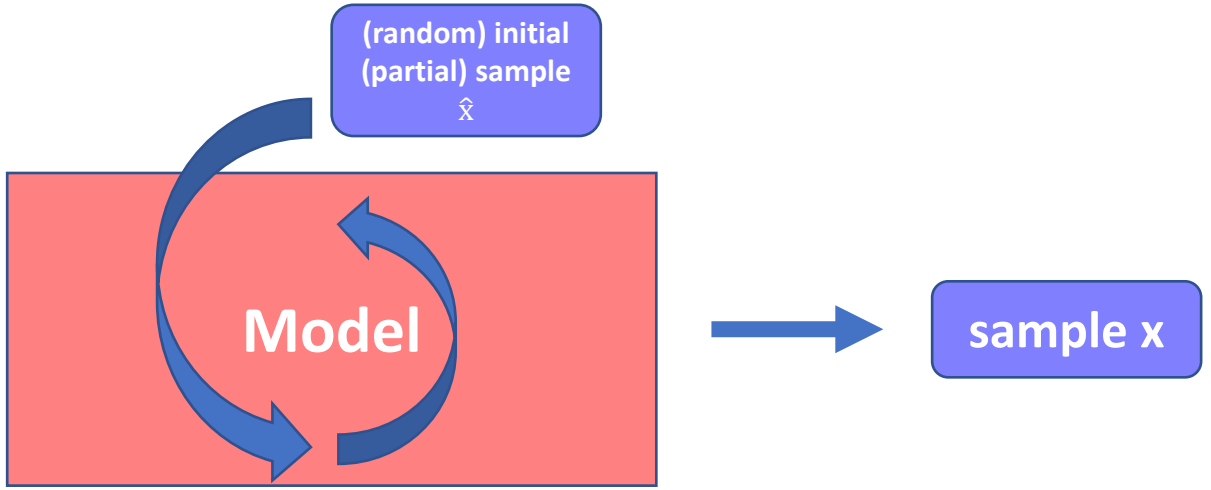


Generative Models: Goals

1st goal: Sample generation

- Generate realistic samples (from the same distribution as training data, *i.e.*, P(**x**))
  - latent variable models
  - fully-observable models

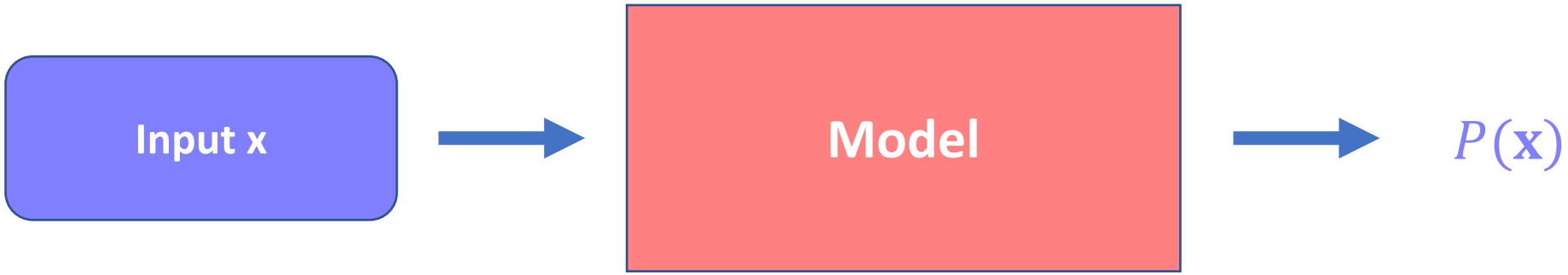
- Assign likelihood to samples



Generative Models: Goals

2nd goal: Density Estimation

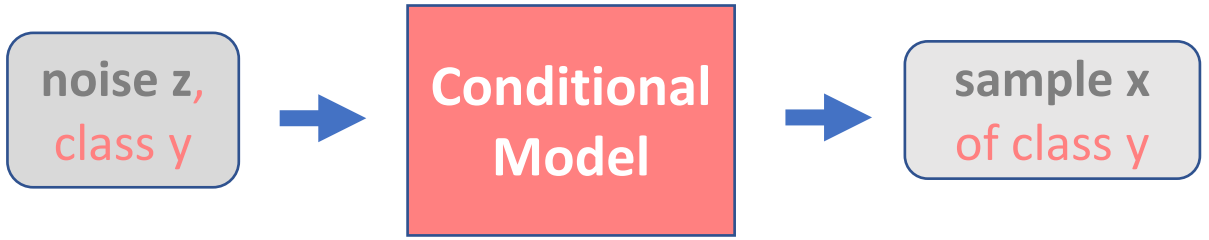
- Generate realistic samples
- Assign likelihood to samples (density Estimation)



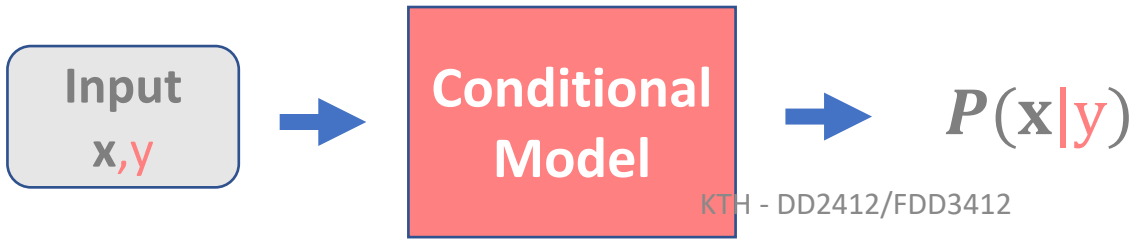
Generative Models: Goals

Conditional generative models' goals

- Generate realistic samples



- Assign likelihood to samples (Density Estimation)



Reflection/Discussion point

- Imagine we are interested in the supervised task of learning $P(y|\mathbf{x})$ on some training data $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$.

- What would be the difference between i) standard discriminative modelling (directly modelling $P(y|\mathbf{x})$) and ii) first learning a conditional generative model $P(\mathbf{x}|y)$, then going from that to $P(y|\mathbf{x})$?

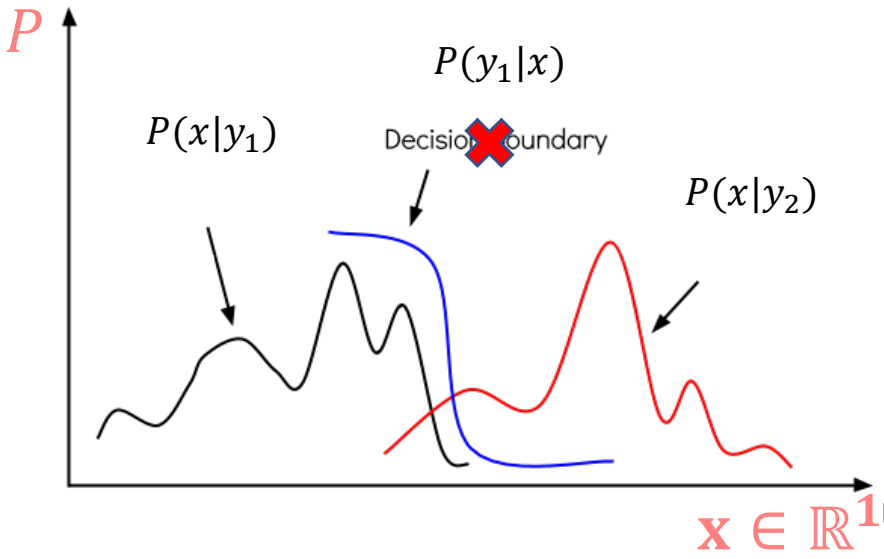
- Why can discriminative modeling be preferable to generative modeling in supervised learning?

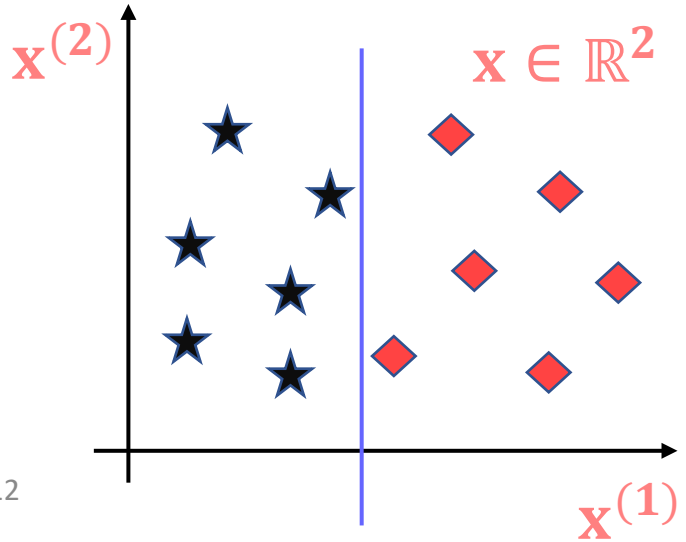


3 minutes!

Discriminative models through generative modeling

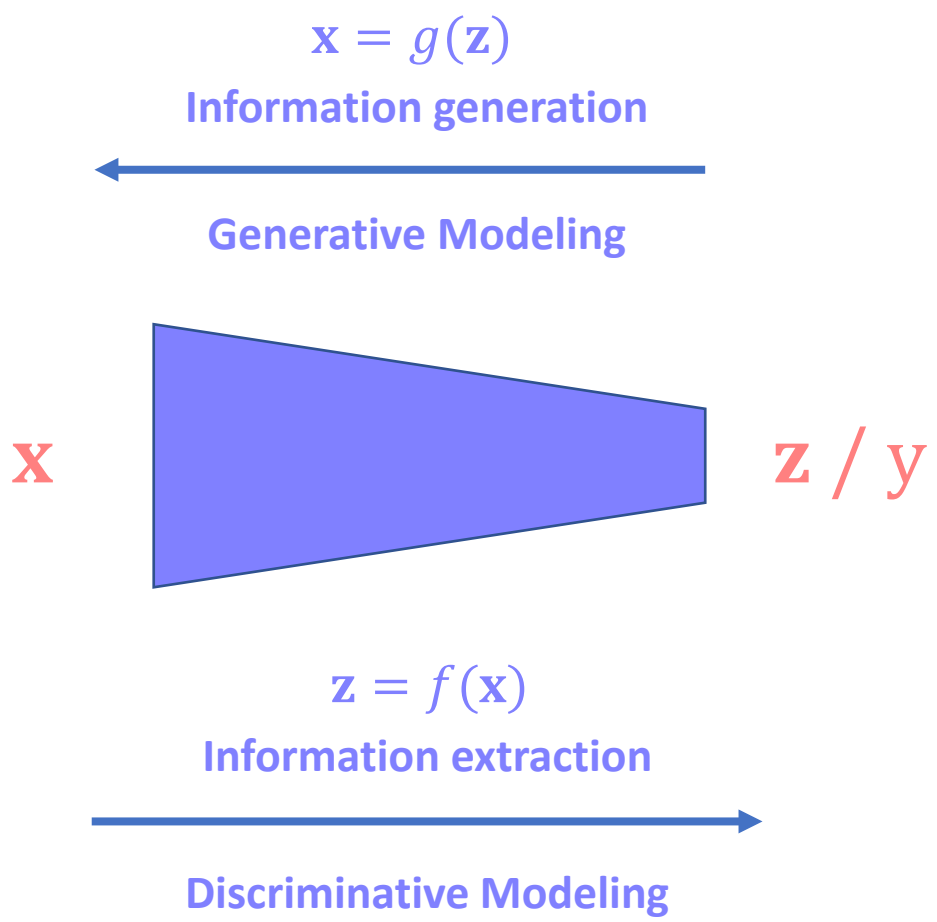
- Generative Models: $P(\mathbf{x})$ or $P(\mathbf{x}|y)$ or maybe $P(\mathbf{x}, y)$
- Discriminative models: $P(y|\mathbf{x})$
  - we can get $P(y|\mathbf{x}) = P(\mathbf{x}, y)/P(\mathbf{x})$ in generative models, especially if we are not interested in normalized probabilities: $P(y|\mathbf{x}) \propto P(\mathbf{x}, y) = P(\mathbf{x}|y)P(y)$





Generative v.s. Discriminative Model

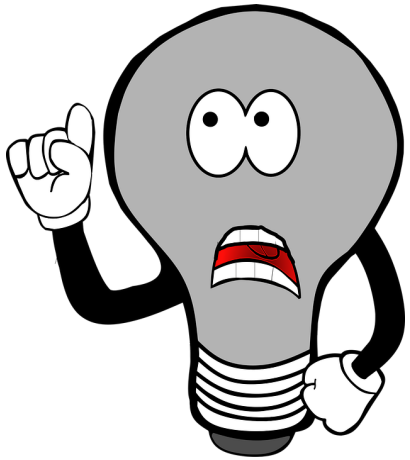
Simple Intuition



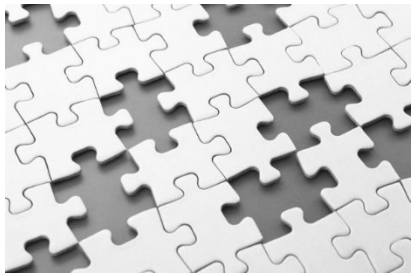
Generative v.s. Discriminative Model

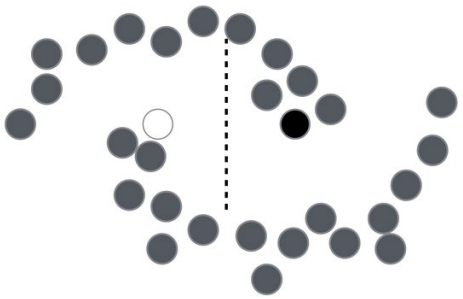
Goods and Bads

- Pros:
  - Out of distribution samples
  - Missing data
  - Missing dimensions
  - Semi-supervised learning
  - Synthetic sample generation
  - …
- Cons
  - Number of data points
  - Number of assumptions











Despite the dimensionality of the input space ($\mathcal{X}$), why can we have a chance to train generative models in lack of enough data?

